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TERMINAL BLOCK, AND I/O UNIT

Abstract

A terminal block includes a plurality of inserted portions and a plurality of release operators, wherein: the inserted portions include first inserted portions and second inserted portions which are arranged in a row in a first direction, and which have mutually different positions in relation to the first direction and a second direction perpendicular to the first direction; and the plurality of release operators are arranged in the first direction, in different positions to the first inserted portions and the second inserted portions in relation to the second direction.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a terminal block and an I/O unit including the terminal block.

BACKGROUND ART

[0002] JP 2007-242486 A discloses a terminal block (terminal block device). The terminal block is attached to an electrical device. The terminal block includes a plurality of insertion-receiving portions (terminals) and a plurality of release operating elements (push buttons). The plurality of release operating elements correspond to different insertion-receiving portions.

SUMMARY OF THE INVENTION

[0003] All the insertion-receiving portions according to JP 2007-242486 A are arranged in a row along the height direction. Here, it is preferable that the interval between the insertion-receiving portions adjacent to each other in the height direction has a certain length. However, in this case, the terminal block becomes large in the height direction.

[0004] In order to suppress the terminal block from becoming large in the height direction, it is conceivable to arrange the plurality of insertion-receiving portions in two rows. By arranging the plurality of insertion-receiving portions in two rows, the terminal block is prevented from being large in the height direction.

[0005] In this case, however, two rows of the plurality of insertion-receiving portions are formed. The two rows are arranged in the width direction orthogonal to the height direction. In other words, the plurality of insertion-receiving portions are arranged in the width direction. As a result, the terminal block becomes large in the width direction.

[0006] In view of the above, there is a problem that it is difficult to configure a small and thin terminal block.

[0007] The present invention has the object of solving the aforementioned problem.

[0008] A first aspect of the invention is a terminal block including: a plurality of insertion-receiving portions into which a plurality of external terminals are inserted, respectively, and which are provided respectively with a plurality of connection terminals each connected to a corresponding one of the inserted external terminals; and a plurality of release operating elements each configured to release connection between a corresponding one of the plurality of external terminals and a corresponding one of the plurality of connection terminals, wherein the plurality of insertion-receiving portions include: a plurality of first insertion-receiving portions arranged in a row along a first direction; and a plurality of second insertion-receiving portions arranged in a row along the first direction at positions that are different from positions of the plurality of first insertion-receiving portions in a second direction orthogonal to the first direction, and wherein the plurality of second insertion-receiving portions are disposed at positions that are different from the positions of the plurality of first insertion-receiving portions in the first direction, and the plurality of release operating elements corresponding to the plurality of first insertion-receiving portions and the plurality of release operating elements corresponding to the plurality of second insertion-receiving portions are arranged along the first direction at positions that are different from the positions of the plurality of first insertion-receiving portions and the positions of the plurality of second insertion-receiving portions in the second direction.

[0009] A second aspect of the invention is characterized by an I/O unit including the terminal block according to the first aspect, wherein the I/O unit has a flat plate shape, and a thickness direction of the flat plate shape is the second direction.

[0010] According to the present invention, it is possible to provide a small and thin terminal block.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1A is a diagram showing a communication system (station) including an I/O unit according to an embodiment; FIG. 1B is a configuration diagram of the I/O unit of FIG. 1A; [0012] FIG. 2 is a front view of a front unit; [0013] FIG. 3A is an end view taken along line IIIA-III A of FIG. 2; FIG. 3B is an end view showing a state where an external terminal is inserted into a first insertion-receiving portion in FIG. 3A; FIG. 3C is an end view showing a state where an operator operates a release operating element in FIG. 3B; [0014] FIG. 4 is an end view taken along line IV-IV of FIG. 2; [0015] FIG. 5A is a front view of a front unit in Comparative Example 1 to which the present embodiment is not applied; FIG. 5B is a front view of a front unit in Comparative Example 2 to which the present embodiment is not applied; FIG. 5C is a front view of a front unit in Comparative Example 3 to which the present embodiment is not applied; [0016] FIG. 6A is a perspective view of a front unit according to a first modification; FIG. 6B is a view showing a plurality of connector units stacked in a height direction; [0017] FIG. 7A is a front view of a front unit according to a second modification; FIG. 7B is a front view of another example of the front unit according to the second modification; and [0018] FIG. 8A is a perspective view of a front unit according to a third modification; FIG. 8B is a diagram schematically showing an internal configuration of a first unit; and FIG. 8C is a diagram schematically showing an internal configuration of a second unit.

DETAILED DESCRIPTION OF THE INVENTION

Embodiments

[0019] FIG. 1A is a diagram illustrating a communication system (station) **11** including an I/O unit (input/output unit) **10** according to an embodiment. FIG. 1B is a configuration diagram of the I/O unit **10** of FIG. 1A.

[0020] A plurality of arrows shown in each drawing indicate directions (height, width, and front-rear) used in the following description. The height direction, the width direction, and the front-rear direction (forward and rearward direction) are orthogonal to each other. The width direction and the front-rear direction are parallel to a horizontal plane. The width direction is a collective term for the L-direction and the R-direction. The L-direction and the R-direction are opposite to each other.

[0021] The communication system **11** is, for example, a system that inputs and outputs signals between a control device and a plurality of devices in order for the control device and the plurality of devices to communicate with each other. The communication system **11** includes a plurality of I/O units **10**. The plurality of I/O units **10** are arranged in the width direction.

[0022] The control device is, for example, a numerical control device that controls a machine tool. The control device is not shown. The device is, for example, an actuator, a detector, or the like provided in the machine tool. The device includes an external terminal **14** and a cable **14a**. The external terminal **14** is a terminal connected to the I/O unit **10**. The cable **14a** is a member that connects the main body of the device and the external terminal **14**. The main body of the device is not shown.

[0023] The communication system **11** may further include an installation member **13**. The installation member **13** is a member that supports the plurality of I/O units **10**. The installation member **13** is, for example, a DIN rail. The extending direction of the installation member **13** illustrated in FIG. 1A is the width direction. That is, the installation member **13** is long in the width direction. In this case, the plurality of I/O units **10** are consecutively arranged on the installation member **13**. The plurality of I/O units **10** are installed in the front portion of the installation member **13**.

[0024] The I/O unit **10** includes a communication processing unit **16**, a base unit **18**, and a front unit **20**. The communication processing unit **16**, the base unit **18**, and the front unit **20** are arranged in the front-rear direction. The communication processing unit **16** is disposed on the rear side of the front unit **20**, and the base unit **18** is disposed on the rear side of the communication processing unit **16**.

[0025] The communication processing unit **16** is a member that houses a control module **22**. The control module **22** is a module that performs processing necessary for establishing communication between the control device and the device. The control module **22** includes, for example, a circuit board with a predetermined electric circuit formed thereon. The electric circuit includes, for example, a control integrated circuit (IC).

[0026] The base unit **18** is a member that supports the communication processing unit **16** in an insertable and removable manner. The insertion direction in which the communication processing unit **16** is inserted into the base unit **18** is the rearward direction. The removal direction in which the communication processing unit **16** is removed from the base unit **18** is the forward direction.

[0027] The base unit **18** includes a plurality of base terminals **24**. The plurality of base terminals **24** are provided on a side portion of the base unit **18**. The plurality of base terminals **24** include base terminals **24L** on a side portion of the base unit **18** that is on the L-direction side and base terminals **24R** on a side portion of the base unit **18** that is on the R-direction side.

[0028] In one I/O unit **10**, the plurality of base terminals **24L** and the plurality of base terminals **24R** are connected to the control module **22** when the communication processing unit **16** is inserted into the base unit **18**. Therefore, the plurality of base terminals **24L** and the plurality of base terminals **24R** are connected to each other via the control module **22** by inserting the communication processing unit **16** into the base unit **18**.

[0029] When two I/O units **10** are adjacent to each other in the width direction, the plurality of base terminals **24R** of the I/O unit **10** on the L-direction side and the plurality of base terminals **24L** of the I/O unit **10** on the R-direction side are connected to each other. Thus, the control modules **22** of the two I/O units **10** adjacent to each other in the width direction are connected to each other.

[0030] Further, the base terminals **24L** of the I/O unit **10** arranged at the most L-direction side among the plurality of I/O units **10** connected in series are connected to the control device. As a result, the control module **22** of that I/O unit **10** is connected to the control device. Further, the control modules **22** of the other I/O units **10** are connected to the control device via the control module **22** of that I/O unit **10**. Accordingly, the device connected to each of the I/O units **10** is connected to the control device via at least one I/O unit **10** (control module **22**).

[0031] However, the plurality of base terminals **24R** of the I/O unit **10** arranged on the most R-direction side among the plurality of I/O units **10** connected in series may be connected to the control device. In this case also, the device connected to each of the I/O units **10** is connected to the control device via at least one I/O unit **10** (control module **22**).

[0032] The front unit **20** is a terminal block that is attachable to and detachable from the communication processing unit **16**. The attaching direction in which the front unit **20** is attached to the communication processing unit **16** is the rearward direction. The detaching direction in which the front unit **20** is detached from the communication processing unit **16** is the forward direction.

[0033] The I/O unit **10** preferably has a flat plate shape in appearance. The thickness direction of the flat plate shape is the width direction. In a case that the I/O unit **10** has a flat plate shape that is thin in the width direction, the plurality of I/O units **10** require less installation space.

[0034] FIG. **2** is a front view of the front unit **20**.

[0035] The front unit **20** will be further described below. In the following description, the height direction is also referred to as a longitudinal direction or a long-side direction. The width direction is also referred to as a transverse direction or a short-side direction.

[0036] The front unit **20** includes a plurality of insertion-receiving portions **26** and a plurality of release operating elements **28**.

[0037] Each of the plurality of insertion-receiving portions **26** is a portion into which the external terminal **14** is inserted. Each of the plurality of insertion-receiving portions **26** has an opening **26a** formed in the front surface of the front unit **20** such that, for example, the external terminal **14** can be fitted into the insertion-receiving portion. However, each of the plurality of insertion-receiving portions **26** is not limited to the opening **26a**. Each of the plurality of insertion-receiving portions **26** may have a mechanism for sandwiching and holding the external terminal **14**.

[0038] The plurality of insertion-receiving portions **26** include a plurality of first insertion-receiving portions **26A** and a plurality of second insertion-receiving portions **26B**. Each of the plurality of first insertion-receiving portions **26A** has a center C.sub.1. Each of the plurality of second insertion-receiving portions **26B** has a center C.sub.2. However, in FIG. 2, illustration of the centers C.sub.1 of some of the first insertion-receiving portions **26A** and the centers C.sub.2 of some of the second insertion-receiving portions **26B** is omitted.

[0039] The plurality of first insertion-receiving portions **26A** are arranged in a row along the height direction. The plurality of second insertion-receiving portions **26B** are also arranged in a row along the height direction. However, the plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B** are disposed at positions different from each other in the width direction.

[0040] The plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B** are also arranged at positions different from each other in the height direction. The center C.sub.2 of each of the second insertion-receiving portions **26B** is located between the centers C.sub.1 of the two first insertion-receiving portions **26A** that sandwich that second insertion-receiving portion **26B**, in the height direction. The plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B** may overlap each other in position in the height direction.

[0041] The plurality of release operating elements **28** are operating elements provided corresponding to the plurality of insertion-receiving portions **26**, respectively. Each of the plurality of release operating elements **28** is, for example, a pusher. Each of the insertion-receiving portions **26** is provided with a connection terminal **30** for connection to the external terminal **14** (see also FIG. 3A). Each of the release operating elements **28** is operated by an operator to release the connection between the corresponding connection terminal **30** and the external terminal **14** connected to that connection terminal **30**.

[0042] The plurality of release operating elements **28** are arranged at positions different from the positions of the plurality of insertion-receiving portions **26** in the width direction. In the present embodiment, all the release operating elements **28** are disposed between the plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B** in the width direction. In the present embodiment, all the release operating elements **28** are arranged in a row along the height direction.

[0043] FIG. 3A is an end view taken along line IIIA-IIIA of FIG. 2.

[0044] Each of the plurality of first insertion-receiving portions **26A** is provided with the connection terminal **30** as described above. The connection terminal **30** is disposed rearward of the opening **26a**, in the first insertion-receiving portion **26A**. The connection terminal **30** is formed into, for example, a plate spring shape and has elasticity. The connection terminal **30** of each of the first insertion-receiving portions **26A** is connected to the control module **22** by the front unit **20** being attached to the communication processing unit **16**.

[0045] FIG. 3B is an end view showing a state where the external terminal **14** is inserted into the first insertion-receiving portion **26A** in FIG. 3A.

[0046] The connection terminal **30** is in contact with the external terminal **14** inserted into the opening **26a**. Thus, the connection terminal **30** and the external terminal **14** are connected to each other. The connection terminal **30** and the external terminal **14** are connected to each other, whereby the device and the control module **22** are connected to each other.

[0047] Further, since the connection terminal **30** has elasticity, the connection terminal **30** can press the external terminal **14** in the width direction. The external terminal **14** is locked into the first insertion-receiving portion **26A** by being pressed in the width direction. In the example of FIG. 3B, the connection terminal **30** presses the external terminal **14** in the L-direction.

[0048] FIG. 3C is an end view showing a state in which the operator operates the release operating element **28** in FIG. 3B.

[0049] The operator can push the release operating element **28** in the rearward direction. By this pushing operation, the release operating element **28** is inserted between the connection terminal **30** and the external terminal **14**. The release operating element **28** is inserted between the connection terminal **30** and the external terminal **14**, and as a result, the contact between the connection terminal **30** and the external terminal **14** is released. When the contact between the connection terminal **30** and the external terminal **14** is released, the pressing of the connection terminal **30** against the external terminal **14** is released. The operator can easily take out the external terminal **14** from the first insertion-receiving portion **26A** by releasing the pressing of the connection terminal **30** against the external terminal **14**.

[0050] FIG. 4 is an end view taken along line IV-IV of FIG. 2.

[0051] The connection terminals **30** are also provided in the second insertion-receiving portions **26B**, respectively. It is preferable that the internal structure of each second insertion-receiving portion **26B** and the internal structure of each first insertion-receiving portion **26A** have structures substantially symmetrical to each other in the width direction with the release operating element **28** as the center (see FIG. 3A and FIG. 4). Thus, the components such as the connection terminal **30** and the release operating element **28** can be configured as common parts for all the insertion-receiving portions **26**.

[0052] The above is a configuration example of the front unit **20**. The positions of the first insertion-receiving portions **26A** in the width direction may be misaligned. That is, the positions of the centers C.sub.1 of the plurality of first insertion-receiving portions **26A** in the width direction may be different from each other. Similarly, the positions of the centers C.sub.2 of the plurality of second insertion-receiving portions **26B** in the width direction may be different from each other.

[0053] FIG. 5A is a front view of a front unit **32A** in Comparative Example 1 to which the present embodiment is not applied.

[0054] In the front unit **32A**, the plurality of insertion-receiving portions **26** and the plurality of release operating elements **28** are alternately arranged in a row along the height direction. The front unit **32A** has at least the following problems.

[0055] A first problem of the front unit **32A** is that it is too large in the height direction. Since the plurality of insertion-receiving portions **26** and the plurality of release operating elements **28** are alternately arranged in a row along the height direction, the front unit **32A** is likely to be larger in the height direction as the number of insertion-receiving portions **26** and the number of release operating elements **28** are larger.

[0056] A second problem of the front unit **32A** is that the operability of the operator is poor. For example, the operator performs an operation of connecting the external terminal **14** to each of the plurality of insertion-receiving portions **26**. In this operation, the external terminal **14** is connected to an insertion-receiving portion **26** that is at a relatively upper position, among the plurality of insertion-receiving portions **26** arranged in the height direction. In this case, the cable **14a** connected to the external terminal **14** droops, and thus the other insertion-receiving portions **26** are hidden behind the cable **14a**. Since the insertion-receiving portions **26** are hidden behind the cable **14a**, it is difficult for the operator to perform the operation of connecting other external terminals **14** to the insertion-receiving portions **26** hidden behind the cable **14a**.

[0057] Further, for example, the operator performs an operation of operating one of the plurality of release operating elements **28**. In this operation, the external terminal **14** is connected to at least one of the two insertion-receiving portions **26** adjacent, in the height direction, to the release operating

element **28** to be operated. In this case, the external terminal **14** may hinder the operator from operating the release operating element **28**.

[0058] The second problem of the front unit **32A** is solved to some extent by widening the interval between the insertion-receiving portion **26** and the release operating element **28** adjacent to each other in the height direction. However, when the interval between the insertion-receiving portion **26** and the release operating element **28** adjacent to each other in the height direction is widened, the first problem described above is further worsened.

[0059] FIG. **5B** is a front view of a front unit **32B** in Comparative Example 2 to which the present embodiment is not applied.

[0060] In order to solve the first problem of the front unit **32A**, it is conceivable to arrange the plurality of insertion-receiving portions **26** and the plurality of release operating elements **28** at positions different from each other in the width direction. That is, the front unit **32B** of FIG. **5B** is conceivable. In the front unit **32B**, the plurality of insertion-receiving portions **26** and the plurality of release operating elements **28** are arranged at different positions in the width direction. The front unit **32B** is smaller than the front unit **32A** in the height direction.

[0061] However, the front unit **32B** is still large. That is, in the front unit **32B**, the plurality of insertion-receiving portions **26** are adjacent to each other in the height direction. When the interval between the adjacent insertion-receiving portions **26** is small, there is a concern that operability of the operator may deteriorate. For example, in a state where the external terminal **14** has already been connected to one of the two adjacent insertion-receiving portions **26**, the operator inserts another external terminal **14** into the other of the insertion-receiving portions **26** that is empty. Here, if the interval between the two insertion-receiving portions **26** is small, the external terminal **14** that has already been connected may hinder the operator from doing the operation. In view of this, the interval in the height direction between the insertion-receiving portions **26** adjacent to each other in the height direction needs to be increased to such an extent that the operability of the operator is not deteriorated. As a result, the front unit **32B** can be made smaller than the front unit **32A**, but there is a problem that the front unit **32B** has to be designed to be large in the height direction in consideration of the above-described need to increase the interval.

[0062] FIG. **5C** is a front view of a front unit **32C** in Comparative Example 3 to which the present embodiment is not applied.

[0063] It might be possible to conceive the front unit **32C** in FIG. **5C** as a terminal block that is smaller in the height direction than the front unit **32A** or the front unit **32B**. In the front unit **32C**, the plurality of insertion-receiving portions **26** and the plurality of release operating elements **28** are arranged at positions different from each other in the width direction. In the front unit **32C**, the plurality of insertion-receiving portions **26** are arranged in a plurality of rows. In this case, in a simplistic point of view, the front unit **32C** can be made half or less the height of the front unit **32B**.

[0064] However, if the plurality of insertion-receiving portions **26** are simply arranged in a plurality of rows, as shown in FIG. **5C**, at least two insertion-receiving portions **26** arranged at the same position in the height direction and the two release operating elements **28** corresponding thereto are arranged along the width direction. In this case, the size of the front unit **32C** in the width direction is doubled as compared with the front unit **32A** and the front unit **32B**, in a simple point of view. In this example, the number of rows of the plurality of insertion-receiving portions **26** is considered to be two. However, if the number of rows of the plurality of insertion-receiving portions **26** is three or more, the widthwise length of the front unit **32C** is further increased. Such a front unit **32C** is not suitable for the flat plate-shaped I/O unit **10** which is preferably thin, even though it is small in the height direction.

[0065] In this regard, according to the present embodiment, the plurality of insertion-receiving portions **26** are arranged in two rows (see also FIG. **2**). Thus, the front unit **20** can be made smaller in the height direction than the front unit **32A** and the front unit **32B**.

[0066] Further, according to the present embodiment, the plurality of release operating elements **28**

and the plurality of insertion-receiving portions **26** are arranged at different positions in the width direction. Therefore, the release operating elements **28** of the front unit **20** are less likely to be hidden behind the cables **14a**.

[0067] Further, according to the present embodiment, the plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B** are different in position in the height direction. Thus, the plurality of release operating elements **28** corresponding respectively to the first insertion-receiving portions **26A** and the plurality of release operating elements **28** corresponding respectively to the second insertion-receiving portions **26B** can be arranged in a row in the height direction. Thus, the front unit **20** is thinner than the front unit **32C**.

[0068] As described above, the front unit (terminal block) **20** of the present embodiment is suitable for application to a small and thin electronic device (I/O unit **10**), and has good operability for an operator.

Modifications

[0069] Modifications of the above-described embodiment will be described below. However, explanations that overlap with those of the embodiment will be omitted insofar as possible in the following description. Elements that have been described in the above embodiment are denoted by the same reference numerals as in the above embodiment unless otherwise specified.

(Modification 1)

[0070] FIG. **6A** is a perspective view of a front unit **20A** (**20**) according to Modification 1.

[0071] The front unit **20A** includes a plurality of connector units **34**. The plurality of connector units **34** are stacked in the height direction.

[0072] The front unit **20A** may further include a predetermined housing **35**. The housing **35** is a member that houses the stacked connector units **34**. The housing **35** is formed with a plurality of openings **35a** arranged in correspondence with the plurality of insertion-receiving portions **26** and a plurality of openings **35b** arranged in correspondence with the plurality of release operating elements **28**.

[0073] Each opening **35a** exposes the corresponding insertion-receiving portion **26** to the outside of the housing **35** when the plurality of connector units **34** are accommodated in the housing **35**.

When the plurality of connector units **34** are accommodated in the housing **35**, each opening **35b** exposes the corresponding release operating element **28** to the outside of the housing **35**.

[0074] FIG. **6B** is a view showing a plurality of connector units **34** stacked in the height direction.

[0075] The connector unit **34** includes an insertion-receiving portion **26**, a release operating element **28**, and a connection terminal **30**. In each connector unit **34**, the arrangement of the insertion-receiving portion **26**, the release operating element **28**, and the connection terminal **30** is in accordance with FIG. **3A** or FIG. **4**. The axis LA shown in FIG. **6B** passes through the widthwise center of the connector unit **34**. The axis LA is parallel to the front-rear direction.

[0076] In the front unit **20A**, one of two connector units **34** adjacent to each other in the height direction is stacked on the other connector unit **34** in a state of being rotated by 180 degrees about the axis LA parallel to the front-rear direction. That is, the connector unit **34** in which the opening **26a** is positioned on the L-direction side with respect to the release operating element **28** and the connector unit **34** in which the opening **26a** is positioned on the R-direction side with respect to the release operating element **28** are alternately arranged.

[0077] According to the present modification, the insertion-receiving portion **26** of the connector unit **34** whose opening **26a** is positioned on the L-direction side with respect to the release operating element **28**, is, for example, the first insertion-receiving portion **26A**. In this case, the insertion-receiving portion **26** of the connector unit **34** whose opening **26a** is positioned on the R-direction side with respect to the release operating element **28** is the second insertion-receiving portion **26B**.

[0078] That is, according to the present modification, the plurality of first insertion-receiving portions **26A** are arranged in a row in the height direction by stacking the plurality of connector

units **34**. The second insertion-receiving portions **26B** are arranged in a row in the height direction at positions different from the positions of the first insertion-receiving portions **26A** in the widthwise direction. Further, the plurality of release operating elements **28** are arranged in a row in the height direction between the plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B**.

[0079] Therefore, according to the present modification, the front unit **20A** that exhibits the same operational effects as the front unit **20** of the embodiment can be easily manufactured by a simple method of stacking the plurality of connector units **34**.

(Modification 2)

[0080] FIG. **7A** is a front view of a front unit **20B** (**20**) according to Modification 2.

[0081] The order of arrangement of the first insertion-receiving portion **26A**, the second insertion-receiving portion **26B**, and the release operating element **28** in the widthwise direction is not limited to the example of FIG. **2**.

[0082] For example, the plurality of first insertion-receiving portions **26A** may be disposed between the plurality of release operating elements **28** and the plurality of second insertion-receiving portions **26B** in the width direction (FIG. **7A**). Alternatively, the plurality of second insertion-receiving portions **26B** may be disposed between the plurality of release operating elements **28** and the plurality of first insertion-receiving portions **26A** in the width direction.

[0083] FIG. **7B** is a front view of another example of the front unit **20B** (**20**) according to the second modification.

[0084] The positions of the plurality of first insertion-receiving portions **26A** and the plurality of second insertion-receiving portions **26B** in the widthwise direction may overlap. This makes it possible to further reduce the thickness of the front unit **20B** in the width direction.

(Modification 3)

[0085] FIG. **8A** is a perspective view of a front unit **20C** (**20**) according to Modification 3.

[0086] The first modification and the second modification may be combined. For example, the front unit **20C** includes a plurality of connector units **34** (**341**, **342**). The plurality of connector units **34** of the front unit **20C** include a plurality of first units **341** and a plurality of second units **342**. The first units **341** and the second units **342** are alternately stacked. The number of the stacked connector units **34** is not particularly limited.

[0087] The front unit **20C** may further include a housing **35**. The housing **35** houses the plurality of connector units **34** stacked together. The plurality of openings **35a** and the plurality of openings **35b** formed in the housing **35** are appropriately modified according to the configuration of each of the first unit **341** and the second unit **342** described later.

[0088] FIG. **8B** is a diagram schematically showing an internal configuration of the first unit **341**.

FIG. **8C** is a diagram schematically showing an internal configuration of the second unit **342**.

[0089] The first unit **341** includes an insertion-receiving portion **26**, a release operating element **281** (**28**), and a connection terminal **30**. The second unit **342** includes an insertion-receiving portion **26**, a release operating element **282** (**28**), and a connection terminal **30**.

[0090] The first unit **341** and the second unit **342** are different from each other in the widthwise position (position in the width direction) of the opening **26a** of the insertion-receiving portion **26**. Accordingly, the first unit **341** and the second unit **342** may be different from each other in the widthwise position of the connection terminal **30**.

[0091] Therefore, when the plurality of first units **341** and the plurality of second units **342** are stacked, the widthwise position of the opening **26a** of each first unit **341** is different from the widthwise position of the opening **26a** of each second unit **342**.

[0092] The plurality of release operating elements **281** and the plurality of release operating elements **282** are arranged on the L-direction side or the R-direction side relative to the opening **26a**. When each release operating element **281** is disposed on the L-direction side of the opening **26a**, each release operating element **282** is also disposed on the L-direction side of the opening **26a**.

When each of the release operating elements **281** is disposed on the R-direction side of the opening **26a**, each of the release operating elements **282** is also disposed on the R-direction side of the opening **26a**.

[0093] Therefore, when the plurality of first units **341** and the plurality of second units **342** are stacked, the plurality of release operating elements **281** and the plurality of release operating elements **282** are arranged in a row in the height direction.

[0094] Each of the release operating elements **281** includes a first portion **36** and a second portion **38**.

[0095] The first portion **36** is a portion extending in the width direction. The first portion **36** has an end **36a** closer to the opening **26a** and an end **36b** further from the opening **26a**. Here, at least the end **36b** of the first portion **36** is exposed to the outside so that the operator can perform a pushing operation. The first portion **36** moves in the rearward direction in response to the pushing operation by the operator. The first portion **36** moves in the forward direction in response to the release of the pushing operation by the operator.

[0096] The second portion **38** is a portion for operating the connection terminal **30** in response to the operator's pushing operation of the first portion **36**. That is, similarly to the release operating element **28** in FIG. 3C, the second portion **38** is a portion for being inserted between the connection terminal **30** and the external terminal **14** to thereby release the connection between the connection terminal **30** and the external terminal **14**.

[0097] The second portion **38** extends from the first portion **36** in the rearward direction. The second portion **38** moves in the rearward direction together with the first portion **36** in response to the first portion **36** being pushed. The second portion **38** moves in the forward direction together with the first portion **36** in response to the release of the pushing operation of the first portion **36**.

[0098] The release operating element **282** also includes a first portion **36** and a second portion **38**. However, the first portion **36** of the release operating element **282** and the first portion **36** of the release operating element **281** have different lengths in the width direction in accordance with the difference in the position of the connection terminal **30** in the width direction between the first unit **341** and the second unit **342**. That is, the first portion **36** illustrated in FIG. 8C is longer in the width direction than the first portion **36** illustrated in FIG. 8B.

[0099] The front unit **20C** can be easily manufactured by alternately stacking the first units **341** and the second units **342** in the height direction.

(Combination of Plurality of Modifications)

[0100] The above-described modifications may be appropriately combined within a range in which no technical inconsistencies occur.

Invention Obtained from Embodiments

[0101] The inventions that can be grasped from the above-described embodiment and the modifications thereof will be described below.

<First Aspect of Invention>

[0102] A first aspect of the present invention is characterized by the terminal block (**20**) including: the plurality of insertion-receiving portions (**26**) into which the plurality of external terminals (**14**) are inserted, respectively, and which are provided respectively with the plurality of connection terminals (**30**) each connected to a corresponding one of the inserted external terminals; and the plurality of release operating elements (**28**) each configured to release connection between a corresponding one of the plurality of external terminals and a corresponding one of the plurality of connection terminals, wherein the plurality of insertion-receiving portions include: the plurality of first insertion-receiving portions (**26A**) arranged in a row along the first direction; and the plurality of second insertion-receiving portions (**26B**) arranged in a row along the first direction at positions that are different from the positions of the plurality of first insertion-receiving portions in the second direction orthogonal to the first direction, and wherein the plurality of second insertion-receiving portions are disposed at positions that are different from the positions of the plurality of

first insertion-receiving portions in the first direction, and the plurality of release operating elements corresponding to the plurality of first insertion-receiving portions and the plurality of release operating elements corresponding to the plurality of second insertion-receiving portions are arranged along the first direction at positions that are different from the positions of the plurality of first insertion-receiving portions and the positions of the plurality of second insertion-receiving portions in the second direction.

[0103] Thus, it is possible to provide a terminal block suitable for application to a small and thin electronic device.

[0104] The plurality of release operating elements corresponding to the plurality of first insertion-receiving portions and the plurality of release operating elements corresponding to the plurality of second insertion-receiving portions may be disposed between the plurality of first insertion-receiving portions and the plurality of second insertion-receiving portions in the second direction. Thus, the plurality of release operating elements corresponding to the plurality of first insertion-receiving portions and the plurality of release operating elements corresponding to the plurality of second insertion-receiving portions can be arranged substantially in a row, and hence the terminal block can be further reduced in width.

[0105] The terminal block may have a shape in which the first direction is a long-side direction and the second direction is a short-side direction when viewed in a direction orthogonal to the first direction and the second direction. This makes the terminal block more preferable for application to an electronic device that is thin in the width direction.

<Second Aspect of Invention>

[0106] A second aspect of the present invention is characterized by the I/O unit (**10**) including the terminal block according to the first aspect, wherein the I/O unit has a flat plate shape, and the thickness direction of the flat plate shape is the second direction.

[0107] Thus, it is possible to provide a terminal block suitable for application to a small and thin electronic device.

REFERENCE SIGNS LIST

[0108] **10**: I/O unit [0109] **14**: external terminal [0110] **20, 20A to 20C**: front unit (terminal block) [0111] **26**: insertion-receiving portion [0112] **26A**: first insertion-receiving portion [0113] **26B**: second insertion-receiving portion [0114] **28**: release operating element [0115] **30**: connection terminal

Claims

1. A terminal block comprising: a plurality of insertion-receiving portions into which a plurality of external terminals are inserted, respectively, and which are provided respectively with a plurality of connection terminals each connected to a corresponding one of the inserted external terminals; and a plurality of release operating elements each configured to release connection between a corresponding one of the plurality of external terminals and a corresponding one of the plurality of connection terminals, wherein the plurality of insertion-receiving portions include: a plurality of first insertion-receiving portions arranged in a row along a first direction; and a plurality of second insertion-receiving portions arranged in a row along the first direction at positions that are different from positions of the plurality of first insertion-receiving portions in a second direction orthogonal to the first direction, and wherein the plurality of second insertion-receiving portions are disposed at positions that are different from the positions of the plurality of first insertion-receiving portions in the first direction, and the plurality of release operating elements corresponding to the plurality of first insertion-receiving portions and the plurality of release operating elements corresponding to the plurality of second insertion-receiving portions are arranged along the first direction at positions that are different from the positions of the plurality of first insertion-receiving portions and the positions of the plurality of second insertion-receiving portions in the second direction.

2. The terminal block according to claim 1, wherein the plurality of release operating elements corresponding to the plurality of first insertion-receiving portions and the plurality of release operating elements corresponding to the plurality of second insertion-receiving portions are disposed between the plurality of first insertion-receiving portions and the plurality of second insertion-receiving portions in the second direction.
 3. The terminal block according to claim 1, wherein the terminal block has a shape in which the first direction is a long-side direction and the second direction is a short-side direction when viewed in a direction orthogonal to the first direction and the second direction.
 4. An I/O unit comprising the terminal block according to claim 1, wherein the I/O unit has a flat plate shape, and a thickness direction of the flat plate shape is the second direction.
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